LESSON 5.2
SOLVING INEQUALITIES – PART 1

LESSON INTRODUCTION

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

LEARNING TARGETS

- I can solve inequalities and represent the solutions on a number line.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.

WORKING TOWARD

MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.

ESSENTIAL QUESTIONS

- How do you solve a two-step inequality?
- How do you represent the solution of a two-step inequality on a number line?

LESSON NARRATIVE

In this lesson, students will learn how to solve two-step inequalities algebraically, including two-step inequalities that employ the distributive property. Students examine common errors and verbalize the error as well as correct the error.

COMPONENT SUMMARY

5.2.1 WARM-UP *(~5 MIN.)*

Solve two-step inequalities

5.2.2 GUIDED INSTRUCTION *(10 MIN.)*

Discuss different ways to solve a two-step inequality

5.2.3 GUIDED PRACTICE *(10 MIN.)*

Practice solving two-step inequalities

5.2.4 COLLABORATION *(10 MIN.)*

Work with partner to discover errors in given solved problems

5.2.5 YOUR TURN! *(5 MIN.)*

Describe how to solve and solve inequalities

5.2.6 WRAP-UP *(~5 MIN.)*

Reflect on understanding of lesson

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Inequality
- Distributive property

MATERIALS

- (Optional) Whiteboard
- (Optional) Dry-erase markers
- (Optional) Paper

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think that the term next to the parentheses must always be distributed to solve the problem.
- If students are dividing first, they may either try to divide the number by the numbers in the parentheses, or they may forget to divide by numbers on the other side of the inequality.

THINGS TO CONSIDER

- If time is an issue, then consider assigning 5.2.5 Your Turn! for homework.
- For students who misplace or lose a negative, consider having those students use a pen or colored pencil for each negative or subtraction sign.

LITERACY AND LANGUAGE ROUTINES

Clarify, Critique, Correct

Consider having partners write down what they think the error is before discussing to make sure both students are participating in the discussion. Students will discuss with a partner what they believe is incorrect about the given inequalities and how to solve the problems correctly.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

5.2.4 Collaboration provides an opportunity for students to engage in mathematical discourse through analyzing and critiquing the work of others. Use this activity as an opportunity for students to practice analyzing the thinking of others as well as an opportunity to deepen understanding through the critique of presented work.

DIFFERENTIATE INSTRUCTION

Consider allowing students to use whiteboards with dry-erase markers to allow for mistakes to easily be erased and have a bigger visual of the problem. Additional visual entry points can be accessed through the use of graphic organizers, like a Venn diagram. 5.2.2 Guided Instruction requires students to compare how two students solve an inequality. Utilizing a Venn diagram is an effective and clear way for students to compare and contrast each student's methods.

5.2.1 WARM-UP: BELL WORK

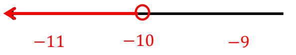
DIFFERENTIATE INSTRUCTION

Use student understanding to inform your instruction and to determine where during this and further lessons. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

5.2.1 Warm-Up: Bell Work

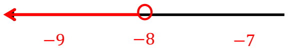
1. Solve and graph each inequality.

   a. $35 < 5 - 3x$

   $x < -10$

   b. $6.5 - \frac{3}{4}n > 12.5$

   $n < -8$

5.2.2 GUIDED INSTRUCTION: SOLVING MULTI-STEP INEQUALITIES

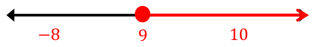
5.2.2 Guided Instruction: Solving Multi-Step Inequalities

Sadie and Promus each solve the inequality $3(x-6) \ge 9$.

Sadie's Work	Promus' Work
$3(x-6) \ge 9$ $3x - 18 \ge 9$ $+18 \quad +18$ $3x \ge 27$ $x \ge 9$	$\frac{3(x-6)}{3} \ge \frac{9}{3}$ $x - 6 \ge 3$ $+6 \quad +6$ $x \ge 9$

1. Compare and contrast Sadie's and Promus' work.
   Sadie distributed the 3 to the numbers in the parentheses while Promus divided both sides by 3 to eliminate the number.

2. Graph the solution to the inequality $3(x-6) \ge 9$.



3. Use substitution to verify the solution set.
   $3(x-6) \ge 9$
   $3(10-6) \ge 9$
   $12 \ge 9$

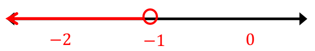
4. Solve the inequality $56 < 7(7-x)$ using Sadie's method and Promus' method.

Sadie's Method	Promus' Method
$56 < 7(7-x)$ $56 < 49 - 7x$ $-49 - 49$ $\underline{7 < -7x}$ $-7 \quad -7$ $-1 > x$ $x < -1$	$\frac{56 < 7(7-x)}{7 \quad 7}$ $8 < 7 - x$ $-7 \ -7$ $\underline{1 < -x}$ $-1 \quad -1$ $-1 > x$ $x < -1$

TEACHER MOVES

Consider providing students with a Venn diagram to compare and contrast Sadie's and Promus's work. Encourage students to discuss how these two students solved the same problem differently but arrived at the same answer. If students are struggling to see the differences, encourage students to circle or highlight the differences in student work to ensure they're the able to compare and contrast the samples effectively. This will provide a visual entry point for learners and encourage students to analyze each piece of work before they compare in words.

5. Graph the solution to the inequality $56 < 7(7-x)$.



6. Use substitution to verify the solution set.
   $56 < 7(7-x)$
   $56 < 7(7-(-3))$
   $56 < 70$

7. Explain whether you prefer Sadie's method or Promus' method.
   Answers will vary. I prefer Sadie's method because I like following order of operations. I prefer Promus' method because I would rather get rid of the 3 in the beginning.

DIFFERENTIATE INSTRUCTION

Consider allowing students to discuss with a partner before writing their answer to help auditory learners synthesize their thinking before writing.

5.2.3 GUIDED PRACTICE: SOLVING INEQUALITIES

DIFFERENTIATE INSTRUCTION

Consider allowing students to use whiteboards with dry-erase markers to allow for mistakes to easily be erased and give students a bigger visual of the problem.

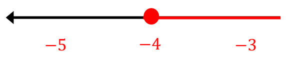
ENRICHMENT

Students may struggle to solve inequalities with negative numbers and/or fractions. If you see students struggling, further scaffold through questioning. When working with a fraction, what could you do instead of dividing by -$\frac{2}{3}$? If you are dividing or multiplying by a negative, what needs to happen to your inequality sign? In question 1d, how can you use the multiplicative inverse of $\frac{1}{8}$ instead of dividing by $\frac{1}{8}$?

5.2.3 Guided Practice: Solving Inequalities

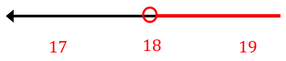
1. Solve each inequality and graph the solution set.

   a. $6(h + 2) \geq -12$

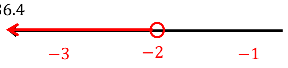
   $h \geq -4$

   b. $-10 > -\frac{2}{3}(p - 3)$

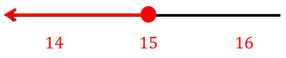
   $p > 18$

   c. $5.6(-4.5 + x) < -36.4$

   $x < -2$

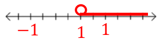
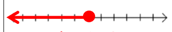
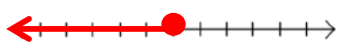
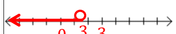
   d. $\frac{3}{4} \geq \frac{1}{8}(\frac{4}{5}b - 6)$

   $b \leq 15$

5.2.4 Collaboration: Error Analysis

1. Work with your partner to identify the errors in the table. Circle the mistake, solve the inequality correctly, then graph the solution set.

Inequality	Correction	Graph
$2(3x-2) > -2$ $6x-4 > -2$ $+4 \quad +4$ $6x > 2$ $-6 \quad -6$ $x > -4$	$2(3x-2) > -2$ $6x-4 > -2$ $+4 \quad +4$ $6x > 2$ $6 \quad 6$ $x > \frac{1}{3}$	
$5-\frac{y}{3} \geq 3$ $-5 \quad -5$ $(-3)\cdot -\frac{y}{3} \geq -2\cdot(-3)$ $y \geq -6$	$5-\frac{y}{3} \geq 3$ $-5 \quad -5$ $(-3) -\frac{y}{3} \geq -2(-3)$ $y \leq 6$	
$2.5-4a \geq 10.5$ $-2.5 \quad -2.5$ $4a \geq 8$ $4 \quad 4$ $a \geq 2$	$2.5-4a \geq 10.5$ $-2.5 \quad -2.5$ $\frac{-4a}{-4} \geq \frac{8}{-4}$ $a \leq -2$	
$\frac{3(2n-3)}{3} < \frac{0}{3}$ $3n-3 < 0$ $+3 \quad +3$ $\frac{3n}{3} < \frac{3}{3}$ $n < 1$	$\frac{3(2n-3)}{3} < \frac{0}{3}$ $2n-3 < 0$ $+3 \quad +3$ $\frac{2n}{2} < \frac{3}{2}$ $n < \frac{3}{2}$	

5.2.4 COLLABORATION: ERROR ANALYSIS

TEACHER MOVES

Consider pulling a small group of students who are struggling to work during the Collaboration. As the errors in the Collaboration are common, discussing these with students who are struggling may help alleviate their misconceptions.

TEACHER MOVES

Clarify, Critique, Correct

Encourage partners to write down or circle what they think the error is before discussing to ensure both students are participating in the discussion and completing the analysis. As students discuss, ensure they are using mathematical language to build on their partner's responses. If necessary, model appropriate clarifications and critiques before students are released to work on their own.

5.2.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider having students who may struggle with writing to perform each step to solve the inequality and label them in order. This will demonstrate understanding of the steps without requiring students to explain in writing, which can be difficult for special populations like ELLs.

5.2.5 Your Turn!

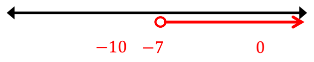
1. Solve the inequality $5(x-1) > -40$ and graph the solution set on a number line.

   $\frac{5(x-1)}{5} > -\frac{40}{5}$

   $x - 1 > -8$

   $\underline{\;+1 \quad +1\;}$

   $x > -7$

2. Describe how to solve the inequality $32 \geq -4(y-10)$.

   Answers will vary. First, divide both sides of the inequality by -4, making sure to reverse the direction of inequality sign. Then, add 10 to both sides of the inequality.

5.2.6 Wrap-Up: Reflection

1. Complete each statement to reflect on today's learning.

   a. When someone asks me what I did in math today, I can say …
   Answers will vary.

   b. The thing that made the most sense to me today was …
   Answers will vary.

   c. Something that I still do not understand is …
   Answers will vary.

5.2.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The information provided from the students here coupled with what was observed during instruction can provide information for differentiating instruction in later lessons.